Jay Collins

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Education

University of Colorado Boulder, Boulder, CO
B.S. Electrical Engineering Expected May 2027
GPA 3.82, Dean's List

Technical Courses

- Circuits I & II (A), Linear Systems (A), Electronics Design Lab (A–), Embedded Software Engineering (A)
- Data Structures (A), Programmable Digital Systems (A), Microelectronics (A), Fields and Waves (A–)
- Calculus I (A–), Calculus II (A), Calculus III (A–), Probability and Statistics (A)

Skills

Languages: C, C++, Python, RISC-V Assembly, Verilog, LaTeX

Protocols & Interfaces: SPI, I2C, UART, I2S, PWM, GPIO Interrupts, Bluetooth, WiFi, REST APIs

Software & Debugging: FreeRTOS, STM32CubeIDE, GDB, UART/Serial Logging, Renode, PyTorch, ExecuTorch, GitHub

Hardware & Testing: PCB Design (Altium), SMD, THT, Soldering, Oscilloscopes, Function Generators, DMM, Power Supplies

Core Engineering Skills: Circuit Design, Embedded Systems, Digital Logic, State Machines, IoT

Professional Skills: Technical Communication, Team Collaboration, Documentation

Experience

Engineering Learning Assistant – CU Boulder Jan 2026 – Present

- Learning Assistant for Linear Systems and Exploring ECE, supporting 150+ students in weekly office hours.
- Explained complex concepts like convolution and Fourier transforms in clear, accessible language.

Deckhand – Hawai'i Nautical May 2024 – Present

- Crew member aboard a 54 ft sailing and snorkeling catamaran, handling bartending, general maintenance, and passenger safety during tours.
- Assisted with maintenance and monitoring of onboard electrical systems, applying hands-on problem-solving to identify and address issues before they impacted operations. Lifeguard and CPR certified.

Projects

Wake-Word Detection on STM32 – Embedded ML

- Built a real-time keyword spotter in C++ on an STM32F407. It runs a 4-class CNN through ExecuTorch on 16 kHz PDM microphone audio, using STFT features and a 1 MB flash budget.
- Trained with time-shift, gain, and noise augmentation and added a tuned voice-activity detector, then validated the system in Renode simulation and on hardware.

World's Worst Alarm Clock – WiFi-Connected Embedded System

- Built an ESP32 alarm clock in C++ with a real-time state machine, interrupt-driven KY-040 encoder/button inputs, SPI SSD1309 OLED refreshing at 30 Hz, I2C DS3231 RTC with NTP sync, and live weather via REST API.
- Moved all networking into a dedicated FreeRTOS task on the ESP32's second core, using task notifications and a mutex to share the radio with Bluetooth audio, so network delays never stall the UI or the alarm.
- Added Bluetooth audio through a MAX98357 I2S amplifier, and took the hardware from breadboard to soldered protoboard to a custom 3D-printed enclosure.

ShapeShifter Hybrid Synthesizer – Senior Capstone (In Progress) Sep 2026 – Apr 2027

- Software and project manager for a 6-person team building a synthesizer for an industry client on a Daisy Seed3 (STM32H750).
- Assembling the full breadboard prototype, with all 16 panel controls wired so far.